Comprobación de ejercicio 6 laboratorio 5

Ejercicio 6 laboratorio 5.

a) $f(x) = x^3 + x^2 - 3x - 5$

Iteración #1 $x_0 = 1$, $x_1 = 1.5$, $x_2 = 2$

Paso 1: $f(x_0) = f(1) = (1)^3 + (1)^2 - 3(1) - 5 = -6$

$f(x_1) = f(1.5) = (1.5)^3 + (1.5)^2 - 3(1.5) - 5 = -3.875$

$f(x_2) = f(2) = (2)^3 + (2)^2 - 3(2) - 5 = 1$

Paso 2: $h_0 = x_1 - x_0 = 1.5 - 1 = 0.5$

$h_1 = x_2 - x_1 = 2 - 1.5 = 0.5$

Paso 3: $\delta_0 = \dfrac{f(x_1) - f(x_0)}{x_1 - x_0} = \dfrac{-3.875 - (-6)}{1.5 - 1} = 4.25$

$\delta_1 = \dfrac{f(x_2) - f(x_1)}{x_2 - x_1} = \dfrac{1 - (-3.875)}{2 - 1.5} = 9.75$

Paso 4: $a = \dfrac{\delta_1 - \delta_0}{h_1 + h_0} = \dfrac{9.75 - 4.25}{0.5 + 0.5} = 5.5$

$b = ah_1 + \delta_1 = 5.5(0.5) + 9.75 = 12.5$

$c = f(x_2) = 1$

Paso 5: $x_3 = x_2 + \dfrac{-2c}{b \pm \sqrt{b^2 - 4ac}}$

$x_3 = 2 + \dfrac{-2(1)}{12.5 \pm \sqrt{12.5^2 - 4(5.5)(1)}}$

$x_3 = 1.9169$

Paso 6: $\varepsilon_a = \left| \dfrac{x_3 - x_2}{x_3} \right| 100\% = \left| \dfrac{1.9169 - 2}{1.9169} \right| 100\% = 4.33\%$

Iteración #2 $x_0 = 1.5$, $x_1 = 2$, $x_2 = 1.9169$

Paso 1: $f(x_0) = f(1.5) = (1.5)^3 + (1.5)^2 - 3(1.5) - 5 = -3.875$

$f(x_1) = f(2) = (2)^3 + (2)^2 - 3(2) - 5 = 1$

$f(x_2) = f(1.9169) = (1.9169)^3 + (1.9169)^2 - 3(1.9169) - 5 = -0.0242$

Paso 2: $h_0 = x_1 - x_0 = 2 - 1.5 = 0.5$

$h_1 = x_2 - x_1 = 1.9169 - 2 = -0.0831$

Paso 3: $\delta_0 = \dfrac{f(x_1) - f(x_0)}{x_1 - x_0} = \dfrac{1 - (-3.875)}{2 - 1.5} = 9.75$

$\delta_1 = \dfrac{f(x_2) - f(x_1)}{x_2 - x_1} = \dfrac{-0.0242 - 1}{1.9169 - 2} = 12.325$

Paso 4: $a = \dfrac{\delta_1 - \delta_0}{h_1 + h_0} = \dfrac{12.325 - 9.75}{-0.0831 + 0.5} = 6.176$

$b = ah_1 + \delta_1 = 6.176(-0.0831) + 12.325 = 11.812$

$c = f(x_2) = -0.0242$

Paso 5: $x_3 = x_2 + \dfrac{-2c}{b \pm \sqrt{b^2 - 4ac}}$

$x_3 = 1.9169 + \dfrac{-2(-0.0242)}{11.812 \pm \sqrt{11.812^2 - 4(6.176)(-0.0242)}}$

$x_3 = 1.9189$

Paso 6: $\varepsilon_a = \left| \dfrac{x_3 - x_2}{x_3} \right| 100\%$

$\varepsilon_a = \left| \dfrac{1.9189 - 1.9169}{1.9189} \right| 100\% = 0.10\%$

b) $f(x) = x^3 - 0.5x^2 + 4x - 3$

Iteración #1 $\quad x_0 = 0, \quad x_1 = 0.5, \quad x_2 = 1$

Paso 1: $f(x_0) = f(0) = (0)^3 - 0.5(0)^2 + 4(0) - 3 = -3$

$f(x_1) = f(0.5) = (0.5)^3 - 0.5(0.5)^2 + 4(0.5) - 3 = -1$

$f(x_2) = f(1) = (1)^3 - 0.5(1)^2 + 4(1) - 3 = 1.5$

Paso 2: $h_0 = x_1 - x_0 = 0.5 - 0 = 0.5$

$h_1 = x_2 - x_1 = 1 - 0.5 = 0.5$

Paso 3: $\delta_0 = \dfrac{f(x_1) - f(x_0)}{x_1 - x_0} = \dfrac{-1 - (-3)}{0.5 - 0} = 4$

$\delta_1 = \dfrac{f(x_2) - f(x_1)}{x_2 - x_1} = \dfrac{1.5 - (-1)}{1 - 0.5} = 5$

Paso 4: $a = \dfrac{\delta_1 - \delta_2}{h_1 + h_0} = \dfrac{5 - 4}{0.5 + 0.5} = 1$

$b = ah_1 + \delta_1 = 1(0.5) + 5 = 5.5$

$c = f(x_2) = 1.5$

Paso 5: $x_3 = x_2 + \dfrac{-2c}{b \pm \sqrt{b^2 - 4ac}}$

$x_3 = 1 + \dfrac{-2(1.5)}{5.5 \pm \sqrt{5.5^2 - 4(1)(1.5)}}$

$x_3 = 0.712$

Paso 6: $\varepsilon_a = \left| \dfrac{x_3 - x_2}{x_3} \right| 100\%$

$\varepsilon_a = \left| \dfrac{0.712 - 1}{0.712} \right| 100\%$

$\varepsilon_a = 40.45\%$

Iteración #2 $x_0 = 0.5$, $x_1 = 1$, $x_2 = 0.712$

Paso 1: $f(x_0) = f(0.5) = (0.5)^3 - 0.5(0.5)^2 + 4(0.5) - 3 = -1$

$f(x_1) = f(1) = (1)^3 - 0.5(1)^2 + 4(1) - 3 = 1.5$

$f(x_2) = f(0.712) = (0.712)^3 - 0.5(0.712)^2 + 4(0.712) - 3 = -0.043$

Paso 2: $h_0 = x_1 - x_0 = 1 - 0.5 = 0.5$

$h_1 = x_2 - x_1 = 0.712 - 1 = -0.288$

Paso 3: $\delta_0 = \dfrac{f(x_1) - f(x_0)}{x_1 - x_0} = \dfrac{1.5 - (-1)}{1 - 0.5} = 5$

$\delta_1 = \dfrac{f(x_2) - f(x_1)}{x_2 - x_1} = \dfrac{-0.043 - 1.5}{0.712 - 1} = 5.36$

Paso 4: $a = \dfrac{\delta_1 - \delta_2}{h_1 + h_0} = \dfrac{5.36 - 5}{-0.288 + 0.5} = 1.698$

$b = a h_1 + \delta_1 = 1.698(-0.288) + 5.36 = 4.87$

$c = f(x_2) = -0.043$

Paso 5: $x_3 = x_2 + \dfrac{-2c}{b \pm \sqrt{b^2 - 4ac}}$

$x_3 = 0.712 + \dfrac{-2(-0.043)}{4.87 \pm \sqrt{4.87^2 - 4(1.698)(-0.043)}}$

$x_3 = 0.7208$

Paso 6: $\varepsilon_a = \left| \dfrac{x_3 - x_2}{x_3} \right| 100\%$

$\varepsilon_a = \left| \dfrac{0.7208 - 0.712}{0.7208} \right| 100\%$

$\varepsilon_a = 1.22\%$